

Automotive 100Base-T1 Physical Layer & SOME/IP Test Report

Signal Quality, BER, and Network Traffic Latency per IEEE 802.3bw

1. 100Base-T1 Physical Layer Bit Error Rate

- Test Case TC-ETH-054: Ethernet BER tested over 5 meters cable length showed zero bit errors ($BER < 10^{-11}$). Full 15-meter harness

2. SOME/IP Transport Protocol End-to-End Latency

- Test Case TC-ETH-056: Ethernet traffic analysis under 70% network load: measured SOME/IP packet latency was 2.8 ms due to queue

3. gPTP Clock Synchronization Simulation

- Clock Drift Prediction: Network simulation in OMNeT++ predicts clock drift synchronization within 420 ns.